

This might not be the right course for you if:

- You're looking for an introduction to machine learning
 - *This course assumes fluency in ML basics (e.g., CS 181 or equivalent), is much more narrow.*
- You're uncomfortable with mathematical abstraction
 - *We'll work with probability measures, duality, metric spaces, and convex optimization.*
- You prefer courses with no theoretical content
 - *Expect definitions, some proofs, and mathematical rigor alongside code and applications.*
- You want *only* theoretical content
 - *We will prove some things, but only as a means to understanding the algorithms.*
- You want a purely software-based class
 - *This is not about "how to use OT libraries" — we'll build algorithms from scratch and understand when and why they work.*
- You're not participating actively
 - *This class includes student-led paper discussion and a research-style final project*

Course Structure

- Format: mix of lectures, guest lectures, and student-led paper discussions

Tues Date	Tuesday	Thurs Date	Thurs
9/2	L1: Intro, Logsitics	9/4	L2: OT Basics, Monge & Kantorovich
9/9	L3: Metric Properties, Dual, Special Cases	9/11	L4: Linear Programming view of OT
9/16	L5: Entropic OT Part I	9/18	L6: Entropic OT Part II
9/23	L7: Semi-Discrete OT	9/25	L8: Dynamic OT Part I
9/30	L9: Dynamic OT Part II	10/2	L10: Connections to Statistical Divergences
10/7	L11: OT as a Loss Function	10/9	Guest Lecture: Kimia Nadjahi
10/14	L12: Extensions	10/16	Guest Lecture: Katy Craig
10/21	Guest Lecture: Remi Flamary	10/23	Guest Lecture: Nicolas Courty
10/28	Student-led paper discussions I	10/30	Student-led paper discussions II
11/4	Guest Lecture: Youssef Mroueh	11/6	Student-led paper discussions III
11/11	Student-led paper discussions IV	11/13	Guest Lecture: Marco Cuturi
11/18	Student-led paper discussions V	11/20	Guest Lecture: Ziv Goldfeld
11/25	Guest Lecture: Pinar Demetci (TBC)	11/27	SPRING BREAK
12/2	Final Project Poster Presentations	12/4	[Reading Period]
12/9	[Reading Period]	12/11	[Finals Period]
12/16	[Finals Period]	12/18	[Finals Period]
	Lecture key: David Guest Discussion		